

Part A: What the tracker reports for one body

$$\begin{aligned}
 F_{\text{trak}}^* &= F_{\text{trak}} \Delta F_{\text{trak}} \\
 &= [R_{\text{trak}}, \vec{p}_{\text{trak}}] [\Delta R_{\text{trak}}, \Delta \vec{p}_{\text{trak}}] \\
 &= [R_{\text{trak}} \Delta R_{\text{trak}}, R_{\text{trak}} \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}}]
 \end{aligned}$$

$$\begin{aligned}
 (F_{\text{trak}}^*)^{-1} &= [(R_{\text{trak}}, \Delta R_{\text{trak}})^{-1}, -(R_{\text{trak}}, \Delta R_{\text{trak}})^{-1} (R_{\text{trak}}, \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}})] \\
 &= [\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1}, -\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} (R_{\text{trak}}, \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}})] \\
 &= [\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1}, -\Delta R_{\text{trak}}^{-1} \Delta \vec{p}_{\text{trak}} - \Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} \vec{p}_{\text{trak}}]
 \end{aligned}$$

$$F_m^* = (F_{\text{trak}}^*)^{-1} F_m \Rightarrow F_m = (F_{\text{trak}}^*)^{-1} F_m^*$$

$$\begin{aligned}
 F_m &= [\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1}, -\Delta R_{\text{trak}}^{-1} \Delta \vec{p}_{\text{trak}} - \Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} \vec{p}_{\text{trak}}] [R_m, \vec{p}_m] \\
 &= [\Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} R_m, \Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} \vec{p}_m - \Delta R_{\text{trak}}^{-1} \Delta \vec{p}_{\text{trak}} - \Delta R_{\text{trak}}^{-1} R_{\text{trak}}^{-1} \vec{p}_{\text{trak}}]
 \end{aligned}$$

Part-B: pointer relative to the reference body

$$F_{ref} = [\Delta R_{tank}^{-1} R_{tank}^{-1} R_{ref}, \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{ref} - \Delta R_{tank}^{-1} \Delta \vec{p}_{tank} - \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{tank}]$$

$$F_{pt} = [\Delta R_{tank}^{-1} R_{tank}^{-1} R_{pt}, \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{pt} - \Delta R_{tank}^{-1} \Delta \vec{p}_{tank} - \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{tank}]$$

$$F_{ref}^{-1} = [R_{ref}, p_{ref}]$$

$$R_{ref} = R_{ref}^{-1} R_{tank} \Delta R_{tank}$$

$$p_{ref} = -R_{ref}^{-1} R_{tank} \Delta R_{tank} (\Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{ref} - \Delta R_{tank}^{-1} \Delta \vec{p}_{tank} - \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{tank})$$

$$= -R_{ref}^{-1} R_{tank} \Delta R_{tank} \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{tank} \Delta R_{tank} \Delta R_{tank}^{-1} \Delta \vec{p}_{tank} + R_{ref}^{-1} R_{tank} \Delta R_{tank} \Delta R_{tank}^{-1} R_{tank}^{-1} \vec{p}_{tank}$$

$$= -R_{ref}^{-1} R_{tank} R_{tank}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{tank} \Delta \vec{p}_{tank} + R_{ref}^{-1} R_{tank} R_{tank}^{-1} \vec{p}_{tank}$$

$$= -R_{ref}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{tank} + R_{ref}^{-1} \vec{p}_{tank}$$

$$F_{ref}^{-1} = [R_{ref}^{-1} R_{task} \Delta R_{task} - R_{ref}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{task} \Delta \vec{p}_{task} + R_{ref}^{-1} \vec{p}_{task}]$$

$$\begin{aligned} F_{ref}^{-1} F_{ptn} &= [(R_{ref}^{-1} R_{task} \Delta R_{task}) (\Delta R_{task}^{-1} R_{task}^{-1} R_{ptn}) + \\ &\quad R_{ref}^{-1} R_{task} \Delta R_{task} (\Delta R_{task}^{-1} R_{task}^{-1} \vec{p}_{ptn} - \Delta R_{task}^{-1} \vec{p}_{task} \\ &\quad - \Delta R_{task}^{-1} R_{task}^{-1} \vec{p}_{task}) - \\ &\quad R_{ref}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{task} \Delta \vec{p}_{task} + R_{ref}^{-1} \vec{p}_{task}] \\ &= [R_{ref}^{-1} R_{ptn}, R_{ref}^{-1} \vec{p}_{ptn} - R_{ref}^{-1} R_{task} \Delta \vec{p}_{task} - R_{ref}^{-1} \vec{p}_{task} \\ &\quad - R_{ref}^{-1} \vec{p}_{ref} + R_{ref}^{-1} R_{task} \Delta \vec{p}_{task} + R_{ref}^{-1} \vec{p}_{task}] \\ &= [R_{ref}^{-1} R_{ptn}, R_{ref}^{-1} \vec{p}_{ptn} - R_{ref}^{-1} \vec{p}_{ref}] \end{aligned}$$

Let b be the tip in the tracker
coordinates

$$\begin{aligned}
 \vec{v} + \Delta \vec{v} &= F_{\text{trak}} \Delta F_{\text{trak}} \vec{b} \\
 &= [R_{\text{trak}} \Delta R_{\text{trak}}, R_{\text{trak}} \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}}] b \\
 &= R_{\text{trak}} \Delta R_{\text{trak}} \vec{b} + R_{\text{trak}} \Delta \vec{p}_{\text{trak}} + \vec{p}_{\text{trak}} \\
 &= R_{\text{trak}} (I + \text{sk}(\alpha_{\text{trak}})) \vec{b} + R_{\text{trak}} \vec{g}_{\text{trak}} + \vec{p}_{\text{trak}} \\
 &= R_{\text{trak}} b + R_{\text{skew}}(\alpha_{\text{trak}}) \vec{b} + R_{\text{trak}} \vec{g}_{\text{trak}} + \vec{p}_{\text{trak}} \\
 \Delta v &= R_{\text{trak}} \text{sk}(\alpha_{\text{trak}}) \vec{b} + R_{\text{trak}} \vec{g}_{\text{trak}} \quad \left\{ v = R_{\text{trak}} \vec{b} + \vec{p}_{\text{trak}} \right\} \\
 &= -R_{\text{trak}} \text{sk}(\vec{b}) \alpha_{\text{trak}} + R_{\text{trak}} \vec{g}_{\text{trak}} \\
 &= \begin{bmatrix} -R_{\text{trak}} \text{sk}(\vec{b}) & R_{\text{trak}} \end{bmatrix} \begin{bmatrix} \alpha_{\text{trak}} \\ \vec{g}_{\text{trak}} \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \|\Delta \vec{v}\| &\leq \|\alpha_{\text{trak}}\| \|\vec{b}\| + \|\vec{g}_{\text{trak}}\| = 0.01 \times 1500 + 5 \\
 &= 20 \text{ mm}
 \end{aligned}$$

This is ten times the 2mm specification,
relative measurement removes it completely.

The body has k markers $1 \dots k$.

a_k - nominal position of marker

b_k - measured position in tracker coordinates

Consider the unknown u the frame
 $F = [R, p]$

If markers were exactly at nominal positions and tracker had no noise.

$$R \vec{a}_k + \vec{p} = \vec{b}_k$$

Residual of marker

$$\begin{aligned} \vec{e}_k &= F \vec{a}_k - \vec{b}_k \\ &= R \vec{a}_k + \vec{p} - \vec{b}_k \end{aligned}$$

Minimize

$$\sum_{k=1}^k \|\vec{e}_k\|^2 = (R \vec{a}_k + \vec{p} - \vec{b}_k)^T (R \vec{a}_k + \vec{p} - \vec{b}_k)$$

$$\begin{aligned} \{ \vec{u}_k = R \vec{a}_k - \vec{b}_k \} &= (\vec{u}_k + \vec{p})^T (\vec{u}_k + \vec{p}) \\ &= \vec{u}_k^T \vec{u}_k + \vec{u}_k^T \vec{p} + \vec{p}^T \vec{u}_k + \vec{p}^T \vec{p} \\ &= \vec{u}^T \vec{u}_k + 2 \vec{u}^T \vec{p} + \vec{p}^T \vec{p} \end{aligned}$$

differentiate w.r.t $\vec{p}$

$$\frac{\partial (\vec{u}^T \vec{u}_k)}{\partial \vec{p}} = 0$$

$$\frac{\partial}{\partial \vec{p}} (2 \vec{u}_k^T \vec{p}) = 2 \vec{u}_k$$

$$\frac{\partial}{\partial \vec{p}} (\vec{p}^T \vec{p}) = 2 \vec{p}$$

$$\begin{aligned} \frac{\partial \|\vec{e}_k\|^2}{\partial \vec{p}} &= 2 \vec{u}_k + 2 \vec{p} \\ &= 2 (R \vec{a}_k - \vec{b}_k + \vec{p}) \end{aligned}$$

Since there are multiple markers

$$\frac{\partial \|\vec{e}_k\|^2}{\partial \vec{p}} = 2 \sum_{k=1}^K (R \vec{a}_k - \vec{b}_k + \vec{p}) = 0$$

$$\Rightarrow R \sum_{k=1}^K \vec{a}_k - \sum_{k=1}^K \vec{b}_k + K \vec{p} = 0$$

$$\begin{aligned} \vec{p} &= \frac{1}{K} \sum_{k=1}^K \vec{b}_k - R \frac{1}{K} \sum_{k=1}^K \vec{a}_k \\ &= \bar{\vec{b}} - R \bar{\vec{a}} \end{aligned}$$

$$\bar{\vec{a}} = \frac{1}{K} \sum_{k=1}^K \vec{a}_k$$

$$\bar{\vec{b}} = \frac{1}{K} \sum_{k=1}^K \vec{b}_k$$

$$\begin{aligned} \vec{e}_k &= R \vec{a}_k + (\bar{\vec{b}} - R \bar{\vec{a}}) - \vec{b}_k \\ &= R(\vec{a}_k - \bar{\vec{a}}) - (\vec{b}_k - \bar{\vec{b}}) \\ &= R \tilde{\vec{a}}_k - \tilde{\vec{b}}_k \end{aligned}$$

$$\begin{aligned} \tilde{\vec{a}}_k &= \vec{a}_k - \bar{\vec{a}} \\ \tilde{\vec{b}}_k &= \vec{b}_k - \bar{\vec{b}} \end{aligned}$$

$$\begin{aligned}
\|e_k\|^2 &= (R\tilde{a}_k - \tilde{b}_k)^T (R\tilde{a}_k - \tilde{b}_k) \\
&= \tilde{a}_k^T R^T R \tilde{a}_k - \tilde{a}_k^T R^T \tilde{b}_k - \tilde{b}_k^T R \tilde{a}_k + \tilde{b}_k^T \tilde{b}_k \\
&= \tilde{a}_k^T \tilde{a}_k - 2 \tilde{b}_k^T R \tilde{a}_k + \tilde{b}_k^T \tilde{b}_k
\end{aligned}$$

Sum over the markers

$$\rightarrow \sum_{k=1}^K \|\tilde{a}_k\|^2 + \sum_{k=1}^K \|\tilde{b}_k\|^2 - 2 \sum_{k=1}^K \tilde{b}_k^T R \tilde{a}_k$$

$$\text{Maximize } \sum_{k=1}^K \tilde{b}_k^T R \tilde{a}_k$$

$$\tilde{b}_k^T (R\tilde{a}_k) = \text{tr}(\tilde{b}_k^T R\tilde{a}_k) = \text{tr}(R\tilde{a}_k \tilde{b}_k^T)$$

$$\sum_{k=1}^K \tilde{b}_k^T R\tilde{a}_k = \sum_{k=1}^K \text{tr}(R\tilde{a}_k \tilde{b}_k^T)$$

$$= \text{tr}\left(R \sum_{k=1}^K \tilde{a}_k \tilde{b}_k^T\right)$$

$$= \text{tr}(RH)$$

$$\{H = \sum_{k=1}^K \tilde{a}_k \tilde{b}_k^T\}$$

$$= \text{tr}(RUSV^T)$$

$$= \text{tr}(SV^T RU)$$

$$= \text{tr}(SM)$$

$$\begin{cases} H = USV^T \\ M = V^T RU \end{cases}$$

M is orthogonal

$$\begin{aligned} M^T M &= (V^T R U)^T (V^T R U) \\ &= U^T R^T V V^T R U \\ &= U^T R^T R U \\ &= U^T U = I \end{aligned}$$

$$\det(SM) = s_1 M_{11} + s_2 M_{22} + s_3 M_{33} \leq s_1 + s_2 + s_3$$

Since $V^T R U = I$

$$V V^T R U U^T = V I U^T$$

$$R = V U^T$$

The bound is reached only when $M = I$

$V U^T$ is orthogonal, so its determinant is ± 1 or -1 . If its -1 , its mirror image.

Flip the sign of direction with singular value to get proper rotation.

$$R = V \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(V U^T) \end{bmatrix} U^T$$

$$\vec{p} = \vec{b} - \left[v \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(VV^T) \end{bmatrix} v^T \right] \vec{a}$$

So now for every tracker observation we can determine a complete rigid transformation.

$$F = \begin{bmatrix} R & \vec{p} \\ 0 & 1 \end{bmatrix}$$

Part C: Finding pointer tip

$$F_{rel} = F_{rel}^{-1} F_{ptk}$$

$$\vec{p}_{tip}^{rel} = R_{rel} \vec{p}_{tip} + p_{rel}$$

Unknown point fixed to the pointer - $\vec{p}_{tip}$

$$\vec{b}_{tip} = R_{ptk} \vec{p}_{tip} + \vec{p}_{ptk}$$

$$\left\{ \begin{array}{l} \text{same as} \\ b_k = R \vec{a}_k + \vec{p} \end{array} \right\}$$

When surgeon pivots the pointer around the calibration sensor, the tip remains at one physical location

fixed point - $\vec{p}_{pivot}$

$$R_i \vec{p}_{tip} + \vec{p}_i = \vec{p}_{pivot} \quad (\text{observation } i)$$

$$R_j \vec{p}_{tip} + \vec{p}_j = \vec{p}_{pivot}$$

Group terms

$$(R_i - R_j) \vec{p}_{tip} + (\vec{p}_i - \vec{p}_j) = 0$$

$$(R_i - R_j) \vec{p}_{tip} = \vec{p}_j - \vec{p}_i \quad (\text{calibration equation})$$

$$\vec{P}_{tip} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix}$$

what does one observation do?

$$(R_i - R_1) \vec{P}_{tip} = \vec{P}_1 - \vec{P}_i$$

$$(R_i - R_1) \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} P_{1x} - P_{ix} \\ P_{1y} - P_{iy} \\ P_{1z} - P_{iz} \end{bmatrix}$$

$$a_{11} P_x + a_{12} P_y + a_{13} P_z = P_{1x} - P_{ix}$$

$$a_{21} P_x + a_{22} P_y + a_{23} P_z = P_{1y} - P_{iy}$$

$$a_{31} P_x + a_{32} P_y + a_{33} P_z = P_{1z} - P_{iz}$$

One pair of pointer observations gives us three equations involving three unknown components of tip

But this isn't enough. It's not about the number of equations but about the independent information the equations contain.

Eg. Let's say the pointer barely changes orientation, $R_i \approx R_1$

$$R_i - R_1 \approx 0$$

our equation becomes

$$0 \cdot \vec{p}_{tip} \approx \vec{p}_1 - \vec{p}_i$$

That leaves us with no information about $\vec{p}_{tip}$

We want the rotational diversity between observations

Different pointer observations provide the information needed to determine $\vec{p}_{tip}$

So we add more observations
Using observation 1 as our reference

$$(R_2 - R_1) \vec{p}_{tip} = \vec{p}_1 - \vec{p}_2$$

$$(R_N - R_1) \vec{p}_{tip} = \vec{p}_1 - \vec{p}_N$$

we now have many equations for
the same unknown vector

Stack the

$$M_i = R_i - R_1$$

$$\vec{b}_i = \vec{p}_1 - \vec{p}_i$$

$$M_i \vec{p}_{tip} = \vec{b}_i$$

$$M_i = \begin{bmatrix} R_2 - R_1 \\ \vdots \\ R_N - R_1 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} \vec{p}_1 - \vec{p}_2 \\ \vdots \\ \vec{p}_1 - \vec{p}_N \end{bmatrix}$$

$$M \in \mathbb{R}^{3(N-1) \times 3}$$

$$b \in \mathbb{R}^{3(N-1)}$$

$$\mathbb{R}^{87 \times 3}$$

{ for 30 observations }

Minimizing

$$\vec{e} = M \vec{p}_{tip} - \vec{b}$$

$$J(\vec{p}_{tip}) = \| M \vec{p}_{tip} - \vec{b} \|^2$$

$$\hat{\vec{p}}_{tip} = \arg \min_{\vec{p}_{tip}} \| M \vec{p}_{tip} - \vec{b} \|^2$$

$$\begin{aligned} J(\vec{p}_{tip}) &= (M \vec{p}_{tip} - \vec{b})^T (M \vec{p}_{tip} - \vec{b}) \\ &= (M \vec{p}_{tip})^T (M \vec{p}_{tip}) - (M \vec{p}_{tip})^T \vec{b} - \vec{b}^T (M \vec{p}_{tip}) + \vec{b}^T \vec{b} \\ &= \vec{p}_{tip}^T M^T M \vec{p}_{tip} - 2 \vec{p}_{tip}^T M^T \vec{b} + \vec{b}^T \vec{b} \end{aligned}$$

$$\frac{\partial J}{\partial \vec{p}_{tip}} = 0$$

$$2 M^T M \vec{p}_{tip} = 2 M^T \vec{b}$$

$$\hat{\vec{p}}_{\text{tip}} = (\mathbf{M}^T \mathbf{M})^{-1} \mathbf{M}^T \mathbf{b}$$

Ideal pivot equation

$$t_{\text{eff},i} = \vec{p}_{\text{sensed},i} - \vec{p}_i$$

$$\vec{p}_{\text{sensed},i} = R_i \vec{p}_{\text{tip}} + \vec{p}_i$$

$$t_{\text{eff},i} = \vec{p}_{\text{sensed},i} - \vec{p}_i = R_i \vec{p}_{\text{tip}}$$

$$R_i \vec{p}_{\text{tip}} - R_1 \vec{p}_{\text{tip}} = t_{\text{eff},i} - t_{\text{eff},1}$$

$$(R_i - R_1) \vec{p}_{\text{tip}} = t_{\text{eff},i} - t_{\text{eff},1}$$

We now know that we at least need 2 observations and our experiment uses 30 observations and provides many more equations than unknowns, while different orientations make the system overdetermined.

Suppose the ideal relationship is

$$M \vec{p}_{\text{tip}} = b_{\text{true}}$$

$$M \vec{p}_{\text{tip}} = b - \epsilon$$

$$\begin{cases} b = b_{\text{true}} + \epsilon \\ b_{\text{true}} = b - \epsilon \end{cases}$$

Least square estimate

$$\hat{\vec{p}}_{\text{tip}} = (M^T M)^{-1} M^T b$$

$$= (M^T M)^{-1} M^T (M \vec{p}_{\text{tip}} + \epsilon)$$

$$= (M^T M)^{-1} M^T M \vec{p}_{\text{tip}} + (M^T M)^{-1} M^T \epsilon$$

$$\hat{\vec{p}}_{\text{tip}} = \vec{p}_{\text{tip}} + (M^T M)^{-1} M^T \epsilon$$

$$\hat{\vec{p}}_{\text{tip}} - \vec{p}_{\text{tip}} = (M^T M)^{-1} M^T \epsilon$$

Let's take on assume $\text{Cov}(\epsilon) = \sigma^2 I$

$$\hat{\vec{p}}_{\text{tip}} - \vec{p}_{\text{tip}} = (M)^+ \epsilon$$

$$\text{Cov}((M)^+ \epsilon) = M^+ \text{Cov}(\epsilon) (M^+)^T$$

$$= M^+ (\sigma^2 I) (M^+)^T$$

$$= \sigma^2 (M^+ (M^+)^T)$$

$$= \sigma^2 ((M^T M)^{-1} M^T M (M^T M)^{-1})$$

$$= \sigma^2 (M^T M)^{-1}$$

$$\text{Cov}(\hat{p}_{\text{tip}}) = \sigma^2 (M^T M)^{-1}$$

from residual

$$e = M \hat{p}_{\text{tip}} - b$$

there are $m = 3(N-1)$ scalar equations
in our stacked system.

we estimated three unknown quantities
 $\text{dof} = m - 3$

$$\sigma^2 = \frac{e^T e}{m-3}$$

$$\begin{aligned} \text{Cov}(\hat{p}_{\text{tip}}) &= \frac{e^T e}{m-3} (M^T M)^{-1} \\ &= \frac{e^T e}{87} (M^T M)^{-1} \end{aligned}$$

$$\Sigma = \text{Cov}(\hat{p}_{\text{tip}})$$

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \Sigma_{22} & \Sigma_{23} \\ \Sigma_{31} & \Sigma_{32} & \Sigma_{33} \end{bmatrix}$$

$$\sum_n = \sigma_x^2$$

$$\sum_{22} = \sigma_y^2$$

$$\sum_{33} = \sigma_z^2$$

So the predicted uncertainty of the calibrated
tip can be $(\sigma_x, \sigma_y, \sigma_z)_{mm}$

Define test error

$$F_{\text{test}} = [R_{\text{test}}, P_{\text{test}}]$$

estimated tip position

$$P_{\text{est}} = R_{\text{test}} \hat{P}_{\text{tip}} + P_{\text{test}}$$

simulation provides the ground truth tip
position

$$P_{\text{true}}$$

Error

$$e_{\text{test}} = \|P_{\text{test}} - P_{\text{true}}\|$$

If we have N test poses, then

$$e_{\text{mean}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} e_{\text{test}, i}$$

maximum error

$$e_{\text{max}} = \max_i e_{\text{test}, i}$$

$$e_{\text{test}} \leq 2 \text{mm}$$